

Simple Speaking Script

What to say for each of the 29 slides, in plain English. Written to be read out loud.

How to use this script

- Each block matches one slide in the deck. Same order.
- *On screen* tells you what the audience is looking at.
- The large text is what you say. You do not have to say it word for word.
- The yellow box is the one number to remember for that slide.
- Short sentences are on purpose. Stop at every full stop and breathe.

Before you start

- Open the web app in a browser tab, on the Calculator page. Slide 21 needs it.
- Speak slowly. Slower than feels normal.
- Say "kilograms of CO₂ per day". Do not say "F". Symbols are for the slide only.
- After you say **9.19 percent** on slide 14, stop for two seconds. Let them think.
- Never say the app is bad. Say the app is correct about the science, and one part needs fixing.
- If the projector fails, you can tell the whole story with four numbers: 62 times, plus or minus 5.92, 9.19 percent, and 0.9997.

The script, slide by slide

Slide 1 Cover — Can We Trust the Eco-Score?

On screen: Title slide. Stay on it while you introduce yourself.

Good morning. My name is Jayed Alam Mansur.

I study Artificial Intelligence at Kathmandu University. I am in year three.

Today I will talk about an app we built. It is called Harit Pathsala.

The app gives a student one score out of 100. The score says how green their daily life is.

My research asks one simple question. Can we trust that score?

Slide 2 One number, used to judge people

On screen: Phone picture on the right, showing the score 41.

We built this app at the Nepal Climate Hackathon in 2025.

It asks a student five questions. How do you travel to school? How much electricity does your home use? How much cooking gas? How much waste do you throw? How much do you spend on pens and notebooks?

Then it gives one number. This student gets 41 out of 100.

Teachers show this number to students. Students compare it with each other. A class dashboard takes the average.

So this number is used to judge people. But nobody had ever checked it. That is why I did this research.

Remember: 41 out of 100

Slide 3

Three problems hidden inside the formula

On screen: Three cards on screen.

Before doing any maths, I looked at the shape of the rule.

The rule is a straight line. If you make 0.5 kilograms of CO₂ in a day, you get 100 points. If you make 3.0 kilograms, you get 0 points.

This creates three problems.

Problem one. Everyone above 3.0 gets the same score: zero.

Problem two. Almost nobody can reach 100.

Problem three. The two limits, 0.5 and 3.0, were simply chosen. They were not calculated from real Nepali homes.

I will show a number for each of these problems later.

Slide 4

Five activities, five constants

On screen: Five coloured rows. Teacher's Question 1 is at the bottom.

First I had to check the numbers inside the app.

Every activity has one fixed number. We call it an emission factor. It tells us how much CO₂ one unit makes.

Travel by bus is 0.016 kilograms of CO₂ for each kilometre. A bus burns a lot of fuel, but forty people share it. So each person carries only a small part.

Electricity is 0.23 kilograms for each unit. Nepal's electricity is mostly hydropower, so this number is low.

One gas cylinder is 21.442 kilograms. That is 14.2 kilograms of gas times 1.51.

Waste is 0.827 kilograms for each kilogram we throw away.

Stationery is a very small number for each rupee we spend.

This answers the first question my teacher asked. I did not change any of these numbers. They come from the hackathon organiser's file. My notebook checks all of them every time it runs. If one number is wrong, the notebook stops and shows an error.

Slide 5

A linear sum, then a clamp

On screen: Two formula boxes. The red box is the important one.

The maths has two steps.

Step one. Multiply each activity by its factor. Then add the five results. This gives kilograms of CO₂ per day.

Step two. Turn those kilograms into a score out of 100.

Step two is where the problem is.

Between 0.5 and 3.0 kilograms, one extra kilogram takes away 40 points. That is good. The score is listening.

But outside that range, one extra kilogram takes away nothing. Zero points. The score has stopped listening.

Please remember this. It is the whole research in one line.

Slide 6

Cooking moves the score 62 times more than stationery

On screen: Five numbers on the left, tornado chart on the right.

Now a useful question. If a student wants a better score, what should they change?

I increased each habit by 10 percent, one at a time, and watched the score.

Cooking: the score drops 2.859 points.

Electricity: 2.76 points.

Waste: 1.985 points.

Travel: only 0.256 points.

Stationery: only 0.046 points.

So cooking is 62 times stronger than stationery.

This means a student can stop buying pens and notebooks for a full year, and the score will move less than one point.

So our advice to students must follow this order. Cooking and electricity first.

Remember: 62 times

Slide 7

In a dead zone, every derivative is exactly zero

On screen: Heatmap on top. Two student boxes at the bottom.

Now the important part. Let us look at two real students.

The first student is very green. She walks to school. Her home has solar panels. She makes about 0.30 kilograms of CO₂ a day. She scores 100.

If she becomes twice as green, her score is still 100. The app cannot see her effort.

The second student is very high. He goes by car. His home uses a lot of electricity. He makes about 4.22 kilograms a day. He scores 0.

If he throws away half as much waste tomorrow, his score changes by zero point zero zero zero.

Not a small change. Exactly zero.

Both students are told their effort is worth nothing. And the app is working exactly as it was designed.

Slide 8

The constants are estimates, not facts

On screen: Teacher's Question 2 at the top. Five percentage boxes at the bottom.

My teacher asked: which model did we use before, and why Monte Carlo now?

Before, the model was fixed. We treated every emission factor as an exact number. Same answers in, same score out, every time.

But those factors are not exact. The electricity factor, 0.23, is not like pi. It is an estimate. Nepal Electricity Authority calculates it again every year.

When there is more rain, we use more hydropower and the number goes down. When there is less rain, it goes up. Across their reports from 2018 to 2023, it moves about 15 percent.

Waste changes about 20 percent between homes. The stationery factor is the least certain, about 25 percent.

So now each factor is not one number. It is a range. And the question changes. Not 'what is the score', but 'how well do we know the score'.

Slide 9

Why not one of these instead?

On screen: Teacher's Question 3, then a table of four methods.

My teacher also asked: what other methods could we have used?

I looked at four other methods before choosing.

Interval arithmetic. This takes the worst case for every factor at the same time. That is too pessimistic. It gives a range so wide it is useless.

Latin hypercube sampling. This is a smarter way to pick samples. But we only have five inputs and ten thousand samples. Normal sampling already gives the right answer.

Bootstrap. This method reuses real collected data. We have no real student data yet. So we cannot use it.

Bayesian methods. These need us to guess the shape of every factor first. That is a lot of extra work for very little gain.

So Monte Carlo was the right choice. I will show you exactly why on the next slides.

Slide 10

Same name. Opposite purpose.

On screen: Teacher's Question 4. Two boxes side by side.

My teacher asked a very good question: is this the same Monte Carlo we learn in Design and Analysis of Algorithms?

Same name. Opposite purpose.

In algorithms, the problem has one fixed answer. For example, is 561 a prime number? That answer never changes. Miller-Rabin uses random numbers only to find the answer faster. The randomness is in our method.

In my research, the randomness is in the real world. The electricity factor really does change from year to year. That is nature, not our program.

So I am not guessing a number I could calculate. I am describing a real spread that exists.

The two ideas share a name and a random number generator. Nothing else.

Slide 11**Two methods, one answer**

On screen: Three big numbers, then a chart.

So how precise is one score? I measured it in two different ways.

First, with algebra. Because we only add and multiply, the maths is exact. The answer is plus or minus 5.92 points.

Second, with simulation. I made 10,000 copies of the world. In each copy the factors are slightly different. I scored the same student in every copy. The answer is plus or minus 5.99 points.

The two answers agree within 1.29 percent.

This matters. It means a score of 41 is really somewhere between 29 and 53.

So if two students are ten points apart, we cannot say one is greener. That difference is just noise.

Remember: plus or minus 5.92 points

Slide 12**A score of 7 that shows as 0 in 19.6% of worlds**

On screen: Two charts. Take your time on this slide.

Here is the place where simulation was the only tool that worked.

Take a student at 2.817 kilograms a day. That is just below the 3.0 limit. The app says his score is 7.

But the cut-off is not a smooth line. It is a wall. So part of his possible range goes past 3.0 and gets pushed onto zero.

When I simulate it, 19.6 percent of the possible worlds show his score as zero, not 7.

The algebra cannot tell us this. Only simulation can.

This is why we needed Monte Carlo.

Remember: 19.6% show 0 instead of 7

Slide 13**10,000 students, because there were none**

On screen: Teacher's Question 5, then the distribution table.

My teacher asked: why 10,000 fake students? What happened to the old data?

The honest answer is that there is no old data. We have never collected a proper student survey. There is no dataset that I chose not to use.

So I built a population instead. Ten thousand students.

Synthetic does not mean fake. Every distribution comes from a published Nepali source. Bus is the most common way to travel. The average commute is three kilometres. Electricity comes from the real NEA billing range.

And this is important. I fixed all these numbers **before** I calculated any score. So I could not adjust them to get a nice result.

I used seed 42, so anyone can run it again and get exactly the same students.

But I must be clear about the limit. This cannot prove the emission factors are correct. It can only show how the scoring rule behaves across a whole school.

Slide 14**9.19% of a school is silenced**

On screen: The red bar in the chart. Pause here for two seconds.

Now the main result.

9.19 percent of the school gets exactly zero.

And 0.0 percent reach 100. Not one student out of ten thousand.

Now look closer at the students who got zero. They are not the same as each other.

Their real footprints go from 2.99 kilograms a day up to 8.91 kilograms a day.

That is about 3 times more pollution from one student than another. And the app gives both of them the same number: zero.

And for all of them, any change they make gives zero change in the score.

Remember: 9.19% get exactly zero

Slide 15**Before and after this research**

On screen: Teacher's Question 6, then the comparison table.

My teacher asked for a comparison. What did we know before, and what do we know now?

Before, we treated the emission factors as exact. Now we know their range.

Before, we had no idea how precise a score was. Now we know it is plus or minus 5.92 points.

Before, all five questions looked equally important. Now we know cooking is 62 times stronger than stationery.

Before, nobody had tested the app on a whole school. Now we have tested it on ten thousand students.

And before, we did not know that 9.19 percent of students get no feedback at all.

This table is the summary of my whole project.

Slide 16**The formula is not broken. The clamp is.**

On screen: Two big R-squared numbers, then a chart.

Now the most important question. Is the formula wrong?

I used a regression to test how well the score follows the footprint.

Across all ten thousand students, R squared is 0.8639. That means the score misses about 14 percent of what is happening.

Then I removed only the students stuck at zero, and tested again. R squared becomes 0.9997.

That means for the students the app can actually score, it explains 99.97 percent of the difference between them. It is almost perfect.

So the formula is not broken. The cut-off is the only problem.

And here is why the floor fills up. Electricity, cooking and waste together, before any travel, already average 2.048 kilograms a day. The limit is 3.0. So a normal Nepali student is already 82 percent of the way to zero points, because of decisions at home that they do not control.

Remember: R^2 0.8639 → 0.9997

Slide 17

Replace the clamp. Change nothing else.

On screen: Curve chart on the left, two method boxes on the right.

If the cut-off is the problem, then we change only the cut-off.

We do not touch the emission factors. We do not touch the footprint. We do not change the five questions. Only the last step.

Method A uses the population to set the limits, at 1.404 and 3.227 kilograms. It is better, but it still has walls at both ends.

Method B is my recommendation. It is an S-shaped curve, called a logistic curve. The middle point is 2.1883 kilograms, which is the middle student.

The important thing is that this curve never becomes flat. It gets close to 0 and close to 100, but never touches them.

So there is always somewhere left to go. Every student's effort is seen.

Slide 18

Why the logistic wins

On screen: Comparison table, then the yellow warning box.

Let me compare the three methods.

Students with no feedback. Old way: 9.19 percent. Method A: 10.27 percent, which is worse. Method B: 0.3 percent.

Next, does it keep the correct order? This is the most important row. The old score gives hundreds of students the same zero, so it loses the order. The logistic keeps it perfectly. The correlation is 1.000000.

That means if student A really pollutes less than student B, then A always scores higher. Always.

And where the old score worked well, the new one agrees with it. The correlation is 0.9972. So this is not a replacement. It is an extension.

But I must be honest about the cost. The new score is relative to the population. Fifty means 'a normal student here'. If the population changes, we must fit it again. And we must always show the kilograms next to the score.

Remember: 9.19% → 0.3%

Slide 19

The web application

On screen: Four coloured page cards.

Now let me show you the same research as something you can click.

I built a web app with four pages.

The Calculator scores one student in three ways at the same time.

Research Findings shows the five main results with their charts.

Understanding the Math explains the ideas, with sliders you can move.

About This Project tells the story and has a class simulator.

The app and my notebook share the same maths file. So the app and the paper can never show different numbers.

It is online. Every time I push a change, it updates by itself.

Slide 20

Calculator — one student, three scores

On screen: Screenshot of the Calculator page.

This is the Calculator page.

On the left, the student answers the five questions with sliders.

On the right, three scores appear at the same time. The old score, Method A, and the logistic.

The starting values are the example from our documentation. Bus, two kilometres, ninety units of electricity, one cylinder, medium waste, two hundred rupees. It gives 1.976 kilograms and a score of 41.

This proves the app matches the original app exactly.

Below, the bar chart shows where the pollution comes from. The colours are the same in every chart in my research.

And there is a section that shows how reliable the score is, using the simulation.

Slide 21

Drag the commute from 7 km to 3 km

On screen: The demo table. Do this live in the app if you can.

Now let me show you the problem in a live demo. This is the most important moment.

I set the student to travel by car, seven kilometres.

The old score says zero.

Now watch both numbers while I make the student greener.

Six kilometres. Old score: zero. Five kilometres: zero. Four kilometres: zero. Three kilometres: still zero.

This student cut their travel by more than half. That is a big, real change. And the app told them five times that it was worth nothing.

Now look at the logistic score. Two. Three. Six. Eleven. Eighteen.

It saw every single step.

That is my whole project in one slider.

Remember: Old: 0 every time. New: 2 → 18

Slide 22

Research Findings page

On screen: Screenshot of page two.

This page shows the five findings.

At the top there are four boxes with the main numbers.

Then each finding opens up, with the chart from my notebook inside.

Nothing here is drawn by hand. Every chart comes from the notebook, and every number is read from one results file.

So if I run the analysis again, the app updates too.

Slide 23

Understanding the Math page

On screen: Screenshot of page three.

This page is for someone who wants to know why, not only what.

It explains where each emission factor comes from, with the calculation.

It explains what a derivative is, in simple words. And there is a slider. You move one habit from zero to two hundred percent, and you watch the score go flat when it leaves the working range.

It also answers the Monte Carlo question from my class, with three short paragraphs.

And there is a second slider. You choose the number of simulations, from one hundred to ten thousand. You watch the answer settle down as the number grows. This shows why ten thousand is enough.

Slide 24

About This Project page

On screen: Screenshot of page four.

This page tells the story of the project, and it states the limitations clearly.

It also has a class simulator. You choose a class size, for example forty students. The app creates that class and scores it in all three ways.

So a teacher can see the problem in their own class size, not in a population of ten thousand.

And you can download the class as a spreadsheet.

Slide 25

Five stages before anything reaches a student

On screen: Teacher's Question 7. Five numbered steps.

My last teacher question was: what is the full process after generating the data?

Making the data was the middle of the work, not the end. Five steps remain.

Step one. Collect real answers. I have already written the survey. Seven questions, three minutes, in English and Nepali, and no personal information.

Step two. Fit the scale again using the real numbers. I need at least one hundred students for this.

Step three. Run the whole notebook again and see which results survive. The percentage will change. The flat region will not.

Step four. Change one function in the app. Nothing else.

Step five. Check again every year, because a relative scale moves when the population moves.

Slide 26

One function moves. Everything else stays.

On screen: Green box and orange box.

I want to be very clear about how small this change is.

What stays the same: all five emission factors, the footprint formula, the five questions, the kilograms figure, and the whole existing app and game.

What changes: one function, the one that turns kilograms into a score.

We also order our advice by the real sensitivity, and we show the uncertainty with the score.

So this research does not say the app is bad. It improves one small part of it.

Slide 27

Three questions, three answers

On screen: Three coloured columns.

Let me summarise the research in three parts of mathematics.

First, mathematical modelling. I wrote the app as proper maths. That showed me two things: the derivatives are constant, and the cut-off makes them zero.

Second, sensitivity analysis. This told us which habits matter. Cooking and electricity control the score. Stationery is 62 times weaker.

Third, statistical calibration. This told us the score is only accurate to plus or minus 5.92 points, and that 9.19 percent of students are silenced.

One project, three branches of mathematics.

Slide 28

What I am not claiming

On screen: Three honest limitation boxes.

Before I finish, I want to say clearly what I am **not** claiming.

One. My students are simulated, not real. So 9.19 percent is what these factors produce on a realistic school. It is not a measured fact about a real school. What is solid is the structure: there is a flat region, and real students live in it.

Two. The new score is relative to a population. Fifty means normal for this group. It must be fitted again when the group changes.

Three. I assumed the emission factors move independently. They probably do not. If they move together, the real uncertainty is **larger** than 5.92 points. So my number is a minimum, not a final answer.

I say these openly, because good research shows its own weak points.

Slide 29

Main finding

On screen: Closing slide. Say this slowly, then stop.

So, can we trust the Eco-Score?

Mostly yes. And now we know exactly where it stops being trustworthy, why it happens, how many students it affects, and what to do instead.

The formula was always correct. The boundaries were always the problem.

Thank you. I am happy to take questions.

If you only have three minutes

Use only these four slides, in this order:

Slide 2	One number out of 100. Students are compared with it. Nobody had checked it. <i>(30 seconds)</i>
Slide 14	9.19 percent of a school gets exactly zero. Their real pollution goes from 2.99 to 8.91 kilograms. And their score cannot move. <i>(60 seconds)</i>
Slide 16	Remove those students and the formula explains 99.97 percent. The formula is fine. The cut-off is the problem. <i>(30 seconds)</i>
Slide 21	The live demo. Car, seven kilometres, drag it down to three. Old score stays at zero. New score climbs. <i>(60 seconds)</i>

Questions they may ask

Why not just move the limit from 3.0 to 9.0?

That fixes the bottom but breaks the middle. Our students are mostly between 1.4 and 3.2 kilograms. If we stretch the line to 9, they all bunch together near the top and we cannot tell them apart.

The logistic curve gives us detail where the students actually are, and it still never goes flat.

Are simulated students not just making up the answer?

It would be, if I said 9.19 percent is a fact about a real school. I do not say that.

The distributions come from published sources. I fixed them before I calculated any score, so I could not adjust them. And the mean score of 31.9 is not a comfortable result. I reported it as it came.

What the simulation proves is the structure: a flat region with real students inside it. Change my assumptions and the percentage moves. The flat region stays.

Why is the average score only 31.9? That seems harsh.

Because that is what these emission factors produce. Home electricity, cooking and waste alone already average 2.048 kilograms, and the limit is 3.0.

I did not adjust the numbers to make the average look nicer. The harsh result is the evidence that the limits are set wrong.

Did you change the original app?

No. Not one line. The original file is untouched.

All my work is in a separate folder. And my notebook checks that its maths matches the app exactly, so they can never drift apart.

Is 10,000 students enough?

Yes, and I can show it. In the app there is a slider from one hundred to ten thousand simulations. You can watch the answer settle down.

The error falls with the square root of N . So one hundred times more simulations only gives ten times more accuracy. Ten thousand is the sensible place to stop.

What will you do next?

Collect real data. I need at least one hundred real students before the middle value is stable enough to build a scale on.

Then fit it again, publish the new numbers, and only then suggest changing the live app.

Words to know

If someone asks what a word means, use these short answers.

Emission factor	One fixed number. It says how much CO2 one unit of an activity makes. For example, one kilometre by bus.
Footprint	The total CO2 a person makes in one day, in kilograms.
Eco-Score	The footprint turned into a number out of 100. Higher is greener.
Clamp / cut-off	A hard limit. Any value past the limit is forced back to the limit. This is what makes the score stop moving.
Dead zone	The range where the score does not react at all, however much a student changes.
Derivative	How much the answer changes when we change one input a little. In a dead zone it is exactly zero.
Sensitivity analysis	Testing which inputs actually matter, by changing them one at a time.
Monte Carlo	Running the same calculation many thousands of times with slightly different inputs, to see the range of possible answers.
Standard deviation	How much the answers spread around the middle. We write it as plus or minus.
R squared	How well one number explains another. 1.0 is perfect.
Logistic curve	An S-shaped curve. It gets close to 0 and 100 but never reaches them, so it never goes flat.
Synthetic population	Students created by the computer using realistic patterns, because we have no real survey yet.